Measurement 159 (2020) 107754

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

Model updating of an existing bridge with high-dimensional variables using modified particle swarm optimization and ambient excitation data

Zhiyuan Xia [a,b,c,*], Aiqun Li [c,d], Jianhui Li [e], Huiyuan Shi [a], Maojun Duan [e], Guangpan Zhou [f]

[a] Jiangsu Province Key Laboratory of Structure Engineering, Suzhou University of Science and Technology, Suzhou 215011, PR China
[b] Department of Civil & Environmental Engineering, National University of Singapore, Singapore 117576, Singapore
[c] School of Civil Engineering, Southeast University, Nanjing 211189, PR China
[d] Beijing Advanced Innovation Center for Future Urban Design, Beijing University of Civil Engineering and Architecture, Beijing 100044, PR China
[e] College of Civil Engineering, Nanjing Forestry University, Nanjing 210037, PR China
[f] School of Science, Nanjing University of Science and Technology, Nanjing 210094, PR China

ARTICLE INFO

Article history:
Received 27 November 2019
Received in revised form 11 March 2020
Accepted 13 March 2020
Available online 19 March 2020

Keywords:
Finite element model updating
Modified particle swarm optimization
Gaussian white noise
Existing bridge structure
High dimensional parameter
Vibration excitation test

ABSTRACT

Model updating for bridge engineering structures to obtain precise finite element model is an efficient tool at present. But with the dimension of the potential parameters during optimization increasing, it brings a challenge to present optimization methods. To improve the performance of convergence process and the probability of detecting global extremum, a combination model updating method of modified particle swarm optimization (MPSO) and response surface method was proposed herein. MPSO has less modification to PSO as only modifying the particle position with Gaussian white noise at a probability and the performance was significantly superior than PSO illustrated by testing functions. A beam model updating example based on the proposed method was tested, the convergence ability and accuracy were compared with genetic algorithm which is widely applied in bridge model updating. Finally, the proposed method was successfully applied to model updating of an existing bridge engineering structures with thirteen variables.

1. Introduction

Finite element (FE) model or experimental methods are always used to evaluate the condition or behavior of existing bridge engineering structures [1–3]. The precision of FE analysis is up to the model which is influenced by the variabilities during the service in material properties, geometric dimensions and also the hypotheses applied in modelling process [4]. For experimental methods, the precision is influenced by test equipment and the amount of test items [5,6]. Considering the economic cost and technical feasibility, FE analysis is more suitable for comprehensive assessment of existing bridge structures under multi-conditions. But the shortcomings in precision of FE model must be avoided. Hence, model updating technique was proposed to ensure a precise FE model reflecting the real responses of the structural system [7,8]. Through modifying the variabilities, the results of FE model get closer to the limited experimental behavior of bridge structures [9–11]. Furthermore, with the complexity of recent bridge engineering, more detailed FE model is needed for analysis, where higher dimension of the parameters is considered [1,3]. That means, for model updating of bridge engineering structures, the optimization issue with higher dimensional parameters needs to be solved. The requirements for optimization method were more accuracy, stable and having better global searching ability.

At the early stage, optimization methods like Newton or quasi-Newton were enough to solve model updating of bridge structures [12,13], because the amount of considered stochastic parameters was not high and the relationships between the parameters and interested performances were simple. For engineering optimization issues with higher dimension of parameters and stronger non-linear relationships, traditional established optimization algorithms are easy to fall into local optimal solutions. Hence, an optimization method with higher global searching ability, genetic algorithm (GA) [14], was proposed to be applied in model updating of bridge engineering structures. Meanwhile, to alleviate calculation effort, the response surface method (RSM) [12,13,15] instead of FE model illustrating the complex relationships between the variabilities and responses is combined with optimization meth-

* Corresponding author at: Jiangsu Province Key Laboratory of Structure Engineering, Suzhou University of Science and Technology, Suzhou 215011, PR China.

E-mail addresses: zhiyuanxia@usts.edu.cn (Z. Xia), aiqunli@seu.edu.cn (A. Li), lijianhui@njfu.edu.cn (J. Li), huiyuanshi@usts.edu.cn (H. Shi), dmj@njfu.edu.cn (M. Duan), guangpanzhou@njust.edu.cn (G. Zhou).

https://doi.org/10.1016/j.measurement.2020.107754
0263-2241/

ods. GA learns the ability of duplication and variation from gene to reorganize the values of each updated parameters during iteration steps until the ending criteria arrives, where the changing of parameters is more directional and the convergence ability is more efficient. Deng and Cai [14] applied GA and RSM to realize a model updating of a two-way bridge located in Louisiana with five selected parameters like Young's modules of concrete for deck, girders and diaphragms, density of the deck and stiffness of bearings. Qin et al. [16] used GA and kriging surrogate model to optimize the Caiyuanba Yangtze River Bridge with six variables which were the Young's modules and density of orthotropic steel deck, arch rib and truss girder. Shabbir and Omenzetter [17] optimized a pedestrian cable-stayed bridge with six parameters of vertical and horizontal deck flexural stiffness, deck torsional stiffness, cable axial stiffness, cable tension and bearing stiffness based on improved GA with sequential niche technique which has better global searching ability. But during the application and researching of GA method, the researchers noticed that some meta heuristic optimization methods seemed to have more chance to find global solution with higher dimensional parameters than GA [18,19], especially swarm optimization methods, like particle swarm optimization (PSO), first proposed by Kennedy and Eberhart.

PSO is based on the principles of bionics like the food searching principle of a swarm of birds and the searching process is more targeted. The way is modifying the positions and velocities of birds or particles based on information exchanges of destination (optimal) position. Okasha et al. [20] researched the model updating of a highway I-girders bridge over the Wisconsin River based on PSO method with eighteen physical parameters. Perera et al. [19] compared between the performance of PSO and GA with the model updating of a small RC frame having twelve variables. It was claimed that PSO showed faster convergence and better ability of continuous modifications of potential external solutions. Meanwhile, Vahidi et al. [21] also made the same comparison with a simulated numerical model and the convergence ability of PSO is verified. The fast convergence of standard PSO sometimes led to prematurely ending because of the diversity of particles is limited at the final stage. If the optimization issue has strong nonlinear condition, then the standard PSO is easily trapped in local optimal solutions. To overcome the shortcomings and improve the global searching ability, several modification and application researches had been conducted. Qin et al. [16] combined the GA and PSO method to optimize a truss FE model with six parameters including Young's modules and densities of beams and columns. The advantage of this hybrid method is using the fast convergence of PSO to make the global searching and order the GA to make local searching, which improved the performance of standard PSO. Li et al. [22] combined Niching technique and PSO to realize multi-modal optimization, with the fast convergence ability of PSO and searching ability of Niching, the hybrid method had better global searching ability than standard PSO. Then, Shabbir and Omenzetter [23] optimized the same pedestrian cable-stayed bridge [17] with same six parameters based on aforementioned improved PSO method. That shows the applications of PSOs to model updating have so far been limited. Except the application of PSO to model optimization, PSO is implemented in damage detection and other optimization issues. Nanda et al. [24] used PSO method to detect and quantify the crack with satisfactory precision. Gholizadeh and Moghadas [25] recommended quantum PSO method to optimize the design of structures, where the individual particle was not depicted by velocity and position in mechanic system, but a wave function in the quantum world. Then the appearance of particle was decided by the probability density function depending on the potential area the particle lies in. The essence of above improvements of PSO is increasing the diversity of particles and expand the searching area to avoid prematurely ending. From the aforementioned algorithms, the advantages of PSO applied in high dimensional model updating of bridge structures are revealed, but the deficiencies are exposed. Because the performance of standard PSO is limited and the modifications are complex due to the combination with other algorithms. Considering the self-modification to standard PSO, choosing Gaussian white noise (GWN) as a mutation has the least cost [26] without combining other methods, where every particle is only disturbed by GWN in each iteration step to maintaining the diversity of these particles and increase the searching probability of global optimal solution.

In this paper, the methodology of PSO with Gaussian mutation was introduced into the model updating of bridge structures with high dimensional parameters. Considering the application of RSM of second order polynomial expression to a model updating with twelve parameters [12], the RSM is implemented with the modified PSO to improve the efficiency. Then, the application of the modified PSO to model updating of an existing bridge FE model with 13 variables was analyzed. Section 2 describes the modelling and vibration test of the existing bridge engineering. Section 3 detailly illustrates the comparison of modified PSO algorithm with standard PSO by testing functions, explains the process of optimization applied to engineering model updating issue with RSM and compares the results of numerical example with 10 parameters between modified PSO and GA method to verify the accuracy. Section 4 successfully applies the proposed model updating method to an existing bridge with 13 variables. The conclusions are drawn in Section 5. Considering the explicit RSM expressions of the bridge is tediously long, then the expressions are arranged in Appendix A to provide supplementary data.

2. Modelling and experiment of an existing extra-width bridge

2.1. Description and modelling

The span comprises of double side spans and one main span, which are 53 m and 112 m. The width of the main girder is 52 m, where the ratio of width to main span is up to 1/2.15. The main girder cross-section has a type of double-box with three-cells in each single box and the standard height of the main girder is 2.8 m. The tower has a gate shape with a height of 45 m over the bridge deck. The two cables are arranged at a transverse distance of 31.7 m and the ratio of rise to span is 1/5.28. At the longitudinal direction, the hangers are located at a distance of 5 m. All are shown in Figs. 1 and 2. The construction materials of the girder and tower are both concrete, meanwhile, that of cables and hangers are high strength steel strand.

Considering the width to span ratio is extremely high and the spatial effect is strong. Traditional one- or double- girder modelling

Fig. 1. The scene image.

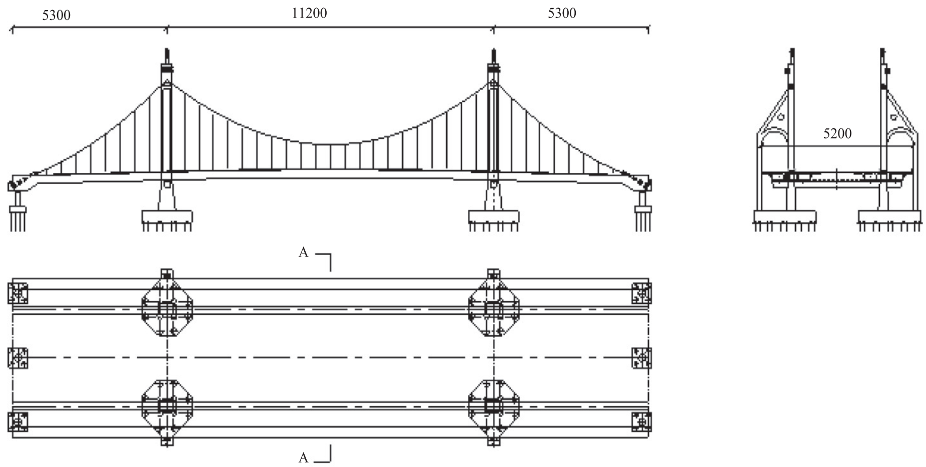


Fig. 2. General layout (unit: cm) by AutoCAD (*i*5-core).

for bridge main girder is not feasible. Herein, the main girder is divided into 8 small girders shown in Fig. 3 during FE modelling and the connection between each two girders is based on the virtual and real transverse beams together with zero-gravity concrete deck. Each partition line coincides with the central line of each cell. The software of ANSYS is introduced into the modelling, where the girders and towers are simulated as 3-D beams, the cables and hangers are simulated as link elements with tension only and all the weights of clamps, saddles and dead load are simulated as mass elements. All the bottom connections between the tower and the foundation are fixed. The connections between the left tower and main girder are releasing only transverse displacement and those between right tower and main girder are releasing both transverse and longitudinal displacements. For the connections between the main girder and foundation at the side span, the bearings on the central axis of the deck are releasing transverse displacement and the others are releasing both transverse and longitudinal displacements. The number of whole elements and nodes of the 3-D model are 20,875 and 15,530 shown in Fig. 4. The characteristics of divided girder are listed in Table 1.

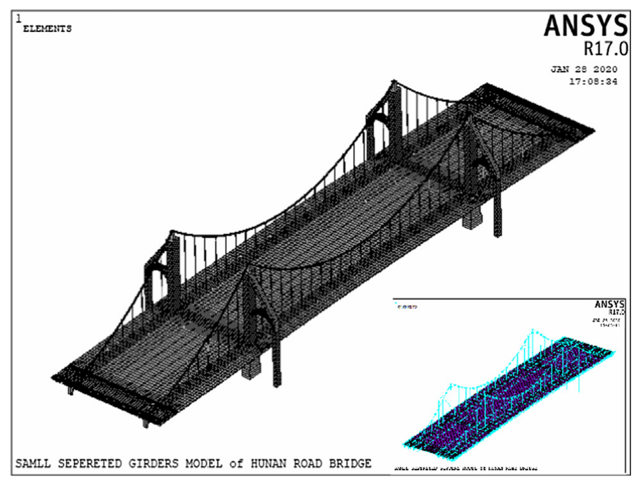


Fig. 4. FE model of the extra-width bridge.

2.2. In-situ experiment and comparisons

To validate the FE model reflecting the dynamical or static responses like displacements and stresses of real bridge structures, an in-situ experiment is usually performed [27–30]. If the limited responses obtained from the test have a good agreement with the model results, then the FE model is proposed to have a relatively acceptable accuracy. Herein, an experiment based on the existing bridge was conducted to obtain the dynamical characteristics of the main girder and several typical vertical displacements under the finished state. The test of the static displacement based on displacement gauge is simple and the mid-span of both side span and main span, quarter-span of main span are selected as the typical points. For the dynamical characteristics test, an ambient excitation vibration was performed and the test points were under each hanger over the deck shown in Fig. 5. Five wireless accelerometers were employed to record the accelerations of each point, where one was acted as reference point and the others were test points. Based on the stochastic subspace identification, the former five dynamical characteristics of the main girder were identified.

The dynamical characteristics of the main girder through vibration test and model analysis are shown in Table 2. The comparisons of the static displacements of typical 5 points at the finished state

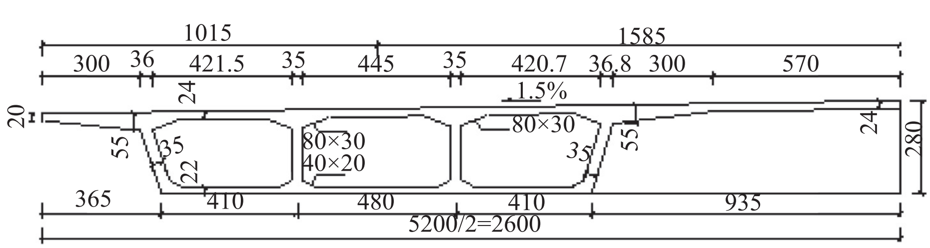
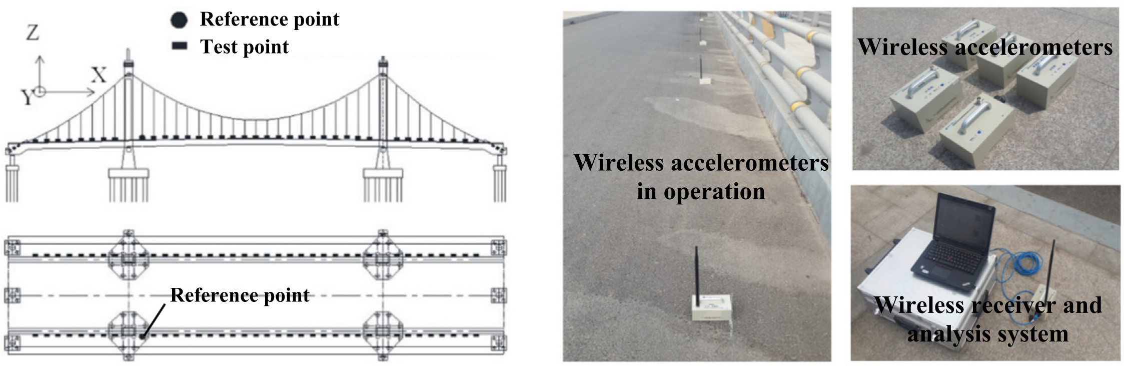


Fig. 3. Layout of semi-section and small girder division (G1, G2, G3 and G4) (unit: cm).

Table 1
Characteristics of divided girders.

Small girder	Area (m^2)	Moment Inertia of vertical bending (m^4)	Moment Inertia of transverse bending (m^4)	Moment Inertia of torsion (m^4)
G1	3.00455	2.45370	5.11041	2.65724
G2, G3	3.18080	3.71211	3.86665	5.77452
G4	4.49056	3.36888	42.89740	2.65728



(a) (b)

Fig. 5. Ambient excitation vibration: (a) Wireless accelerometers arrangement and (b) Recording equipments.

Table 2
Comparisons of dynamical characteristics in the main girder.

Modes	Frequency (Hz)			Error (%)
	Simulation	Test	MAC	
First order of vertical bending	0.7208	0.9030	0.979	25.28
First order of torsion	1.0643	1.3670	0.973	28.44
First order of transverse bending	1.3378	1.3790	0.915	3.08
Second order of vertical bending	1.3892	1.9290	0.894	38.60
Third order of vertical bending	1.7504	2.4170	0.849	38.08

Note: Error = |Simulation-Test|/Simulation × 100%, MAC is modal assurance criteria.

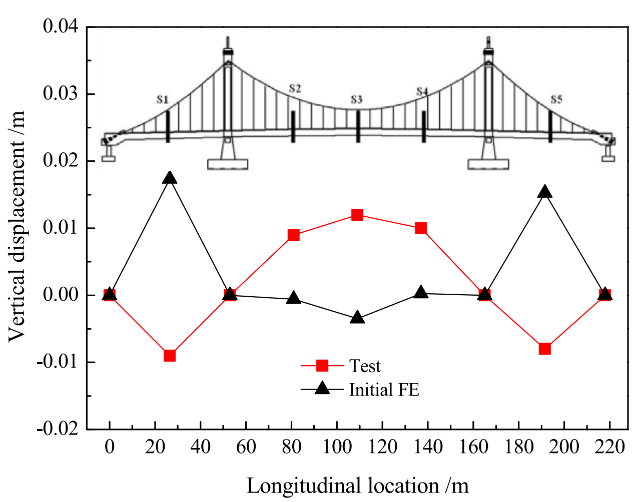


Fig. 6. Vertical displacements of typical points at the finished state.

are shown in Fig. 6. It is obvious that this initial FE model does not reflect the responses of the real structures well, as the differences in the modal frequencies are all serious, even up to 40% and the shape of the girder at the finished state is even opposite between the model and test. Hence, the model updating needs to be employed to solve this problem.

3. Description of modified PSO and model updating process

3.1. Modified PSO

For standard PSO method [19,22–26], the position $\mathbf{X}_i^t$ and velocity $\mathbf{V}_i^t$ of each particle are expressed as

$$\mathbf{X}_i^t = [x_{i1}^t, x_{i2}^t, x_{i3}^t, \cdots, x_{id}^t] \tag{1}$$

$$\mathbf{V}_i^t = [v_{i1}^t, v_{i2}^t, v_{i3}^t, \cdots, v_{id}^t] \tag{2}$$

where the superscript t and subscript i mean the t-th iteration step and the i-th particle respectively, i = 1,2,. . .,N, N is the amount of random particle swarm and d is the dimension of the both position and velocity. For each particle, the position represents a fitness value or objective value $F(\mathbf{X}_i^t)$, where $F(.)$ is the objective function. Then the standard PSO process is explained as following steps. (a) Randomly generate positions and velocities of N particles with a dimension of d; (b) Calculate the values of each position and obtain the objective values of $F(\mathbf{X}_i^t)$; (c) Select the minimum value from $F(\mathbf{X}^1{}_i)$ to $F(\mathbf{X}_i^t)$, whose position is defined as $\mathbf{P}_i^t = [p_{i1}^t, p_{i2}^t, p_{i3}^t, \ldots, p_{id}^t]$ and called the personal best position for the i-th particle until the t-th iteration step; (d) Select the minimum value from $\mathbf{P}^t{}_1$ to $\mathbf{P}_N^t$, whose position is defined as $\mathbf{P}_g^t = [p_{g1}^t, p_{g2}^t, p_{g3}^t, \ldots, p_{gd}^t]$ and called the global best position for the whole particles until the t-th iteration step; (e) Quit the process if $F(\mathbf{P}_g^t)$ meets the stopping criteria or the iteration step t is beyond the maximum step, otherwise, modify the positions and velocities of particles according to Eq. (3) and Eq. (4), then turn back to step (b). The equations are expressed as

$$v_{ij}^{t+1} = \omega v_{ij}^{t} + c_1 r_1 [p_{ij}^{t} - x_{ij}^{t}] + c_2 r_2 [p_{gj}^{t} - x_{ij}^{t}] \tag{3}$$

$$x_{ij}^{t+1} = x_{ij}^{t} + v_{ij}^{t+1} \times \Delta T \tag{4}$$

where coefficient ω is weighting factor, c_i is acceleration coefficient, r_i is random coefficient having an uniformly distribution between [0,1], ΔT is the unit time, i = 1,2,. . .,N, j = 1,2,. . .,d.

When one particle finds a local or global optimal solution, all the particles would gather together close to the former particle through aforementioned iteration steps and the diversity of the particles becomes poor at the later stage. It means the standard PSO has a deficiency of ending prematurely for strong nonlinear optimization issues especially has lots local optimal solutions. The most useful modification is increasing the diversity of the particles through disturbing limited particles to jump out of the aggregation circle and search more optimal solutions. Herein, a modified PSO based on the Gaussian white noise mutation is introduced. The modification to the PSO is not complex, but the improvement in accuracy is significant. In step (e), before modifying the positions of the particles for the next step, select the particles at a probability of P belonged to [0.05,0.15] and mutate the positions of chosen particles through

$$x_{ij}^{t} = p_{gj}^{t} \times (0.5 + \sigma) \tag{5}$$

where σ is a random number from Gaussian white noise.

Three testing functions, F1, F2 and F3 are used to validate the performance of the modified PSO (MPSO). F1 is a single peak function to ensure the efficiency of the MPSO, while F2 and F3 are multi-peak functions to test the accuracy or the global searching ability. All the expressions and figures of two-dimensional issues are shown in Table 3, the minimum values are all zero for those functions.

Based on the PSO and MPSO, optimizations of the above testing functions with ten-dimension are compared. During the process, weighting factor is 0.55, acceleration coefficients are 2, swarm amount is 60, maximum step is 300, the stopping criteria of the error is 0.01 and the selecting probability in MPSO is 0.1. The mean results of the optimal solution and iteration step in 50 times of optimization are shown Table 4 and one-optimization processes for each testing function are compared in Fig. 7. It is obvious that MPSO has a better performance than PSO especially in strong nonlinear optimization issues like F2 and F3. As illustrated before, F2 and F3 are multi-peak functions shown in Table 3 with lots of local extremums. During 50 times optimization processes, MPSO finds the mean global extremum 100% with mean convergence step of 64 and 60 for F2 and F3 respectively. But standard PSO shows bad convergence ability with 0% successful searching. In process of single-peak function F1, although both MPSO and PSO converge to global solution, the iteration step of MPSO is far less than that of PSO and the mean optimal solution is closer to global extremum. Thus, the disturbance of Gaussian white noise to PSO is feasible.

3.2. Surrogate model of RSM

When applying MPSO to model updating of bridge engineering, surrogate model must be employed to reduce the effort of analyzing FE model. During optimization process, the calculation of responses through FE model is replaced by those surrogate models which are explicit expressions between the responses and parameters. The most widely used surrogate model is polynomial RSM and the technique of this model is relative mature. Considering the dimension of parameters is high in engineering structures, the order of polynomial RSM is usually not high. Conventionally, a second-order or a third-order polynomial surrogate model [12] is enough for bridge engineering. Supposing the response $y_k = y_k(\mathbf{X}^{(k)})$ is one of responses of the k-th input parameter group $\mathbf{X}^{(k)}$, then a second-order polynomial surrogate model [12,13,31] is

$$y_k(\mathbf{X}^{(k)}) = \beta_0 + \sum_{i=1}^{d} \beta_{ii} \left(x_i^{(k)}\right)^2 + \sum_{i=1}^{d-1} \sum_{j=i+1}^{d} \beta_{ij} x_i^{(k)} x_j^{(k)} + \sum_{i=1}^{d} \beta_i x_i^{(k)} + \varepsilon_k \tag{6}$$

where d is the dimension of the input parameters, $x_i^{(k)}$ is the i-th parameter among the k-th input parameter vector $\mathbf{X}^{(k)}$, k = 1,2,. . ., N, N is the amount of input parameter vectors, β_0, β_{ii}, β_{ij} and β_i are the regression coefficients; ε_k is the error between the fitting and real value. For instance, if the dimension d is 3 and the number of input parameter vectors k is 10, then the 10 input datasets are

Table 3
Expressions and figures of three testing functions.

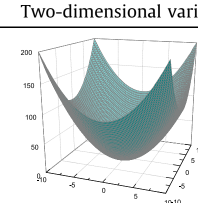
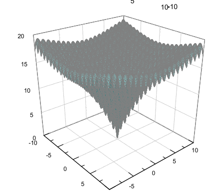
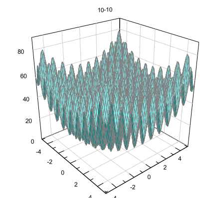
Name	Expression	Variables range	Two-dimensional variables figure
F1	$\min f(x) = \sum_{i=1}^{n} x_i^2$	[−10,10]	
F2	$\min f(x) = -20\exp\left(-0.2\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}\right) - \exp\left(\frac{1}{n}\sum_{i=1}^{n}\cos(2\pi x_i)\right) + 20 + 2.71282$	[−10,10]	
F3	$\min f(x) = \sum_{i=1}^{n}[x_i^2 - 10\cos(2\pi x_i) + 10]$	[−5.12,5.12]	

Table 4
Mean optimization results based on PSO and MPSO.

Functions	Methods	Mean optimal values	Mean iteration step	Successful times/Total times
F1	PSO	0.0008869700	143	50/50
	GMPSO	0.0006203223	29	50/50
F2	PSO	0.5836349220	–	0/50
	GMPSO	−0.0016796473	64	50/50
F3	PSO	8.1046645720	–	0/50
	GMPSO	0.0006176621	60	50/50

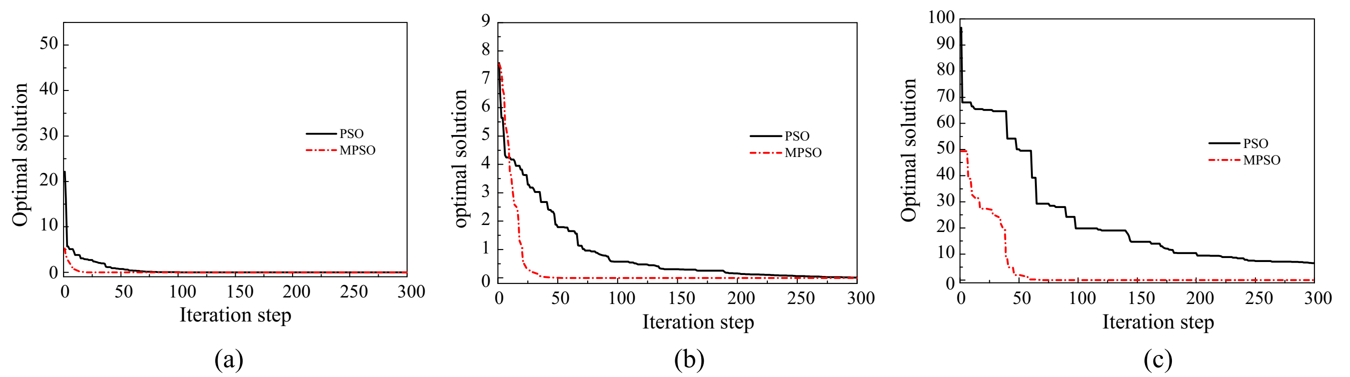


Fig. 7. Optimization processes of three testing functions based on PSO and MPSO: (a) Optimization process of F1, (b) Optimization process of F2, and (c) Optimization process of F3.

obtained, namely $\mathbf{X}^{(k)} = [x_1^{(k)}, x_2^{(k)}, x_3^{(k)}]$, k = 1,2,...,10. In theory, each input parameter vector $\mathbf{X}^{(k)}$ has a corresponding response y_k through FE model, the purpose of RSM is to explore an explicit relationship based on the ten groups of $\mathbf{X}^{(k)}$ and y_k. Hence, the key step is yielding the coefficients of β_0, β_{ii}, β_{ij} and β_i in Eq. (6). β_0 is a constant regression coefficient. β_{ii}s and β_{ij}s are regressions reflecting the influence of the second order items of single parameter x_i and interaction effect of x_i, x_j respectively. β_is are regressions reflecting the influence of the first order items of single parameter x_i. Supposing $\mathbf{X}$ and $\mathbf{Y}$ are the input and output datasets matrix, where $\mathbf{Y}$ is obtained from FE analysis according to parameter matrix $\mathbf{X}$, Eq. (6) is summarized as

$$\mathbf{Y} = \mathbf{B}\mathbf{X} + \mathbf{E} \tag{7}$$

where $\mathbf{B}$ is the regression coefficients matrix including β_0, β_{ii}, β_{ij} and β_i and $\mathbf{E}$ is the error matrix of ε_k. Using the least square method (LSM), the unbiased estimation of $\mathbf{B}$ is

$$\mathbf{b} = \hat{\mathbf{B}} = \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{Y} \tag{8}$$

then the estimation of responses is rewritten as $\mathbf{bX}$ instead of re-running FE analysis.

3.3. Model updating process

The model updating process of MPSO with RSM is similar to that of MPSO and the main task here is recognizing the physical meaning of the alphabets of MPSO connected with the real structures. Herein, the process of model updating is separated by two groups which are second-order polynomial RSM process and model updating process of MPSO. Supposing the updating engineering structure has d parameters to be updated and the optimization object is minimizing the error between the FE model and real structure. Then the steps of model updating are, (a) Generate a number of 15d to 20d parameter vectors with d dimension as input datasets $\mathbf{X}$ based on Latin Hypercube Sampling method [32]; (b) Perform FE model of engineering structure for each parameter vector and obtain the model responses as output datasets $\mathbf{Y}$; (c) Construct the surrogate explicit expressions between the matrix $\mathbf{X}$ and $\mathbf{Y}$ based on Eqs. (6)–(8); (d) Establish the objective function using the error between test responses and surrogate expressions obtained from (c); (e) Start the process of MPSO, where one particle means one parameter vector, the position of the particle means a multi-dimension parameter and the final global best position is the updated parameters. To illustrate this process more clearly and validate the feasibility of the proposed model updating methodology, the model of bridge mentioned in Section 2 with extra-width double-box concrete girder was updated.

3.4. Numerical example

A model updating of a simply supported beam was implemented to illustrate the fast convergence of the proposed method compare with GA method which is widely used in bridge model updating. The span and section of the beam in this numerical example are 10 m and 1 m × 1 m respectively. The beam concrete

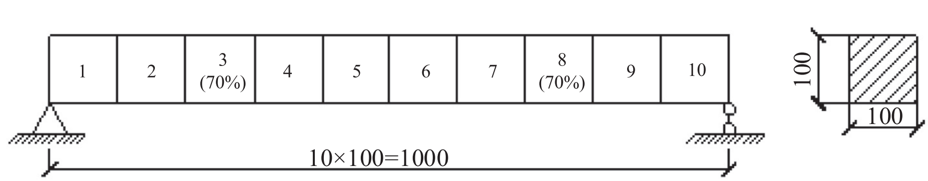


Fig. 8. A numerical beam example (unit: cm).

properties are 3.45×10^{10} Pa of Young's module and 2600 kg/m^3 of density.

A 3-D beam element was used in the FE model of this example based on the software platform of ANSYS, where there was a total of 11 nodes and 10 elements, shown in Fig. 8. The ten variables of vertical flexure stiffness of each element were selected as modified parameters during model updating process. Supposing the vertical flexure stiffnesses only in the third and eighth element had a reduction to 70% of the original stiffness, the purpose of the model updating was optimizing the FE model of the numerical beam with the above ten parameters to make the vibration frequencies of the updated model have an agreement with those of damaged beam model. In this example, the particle in MPSO was described as $\mathbf{X}^{(k)} = [x_1^{(k)}, x_2^{(k)}, \ldots, x_{10}^{(k)}]$ and the searching limitation of each parameter is assumed to be 0.6 to 1.1 times of initial value. Based on the LHS, 200 samples of parameter vectors as input dataset were analyzed to yield the mode frequencies of the beam as output dataset. Then based on the aforementioned fitting process of RSM, the explicit expressions of mode frequencies with the relationship of ten parameters were obtained. The explicit objective function in this example was the minimum of frequency residuals sum of first two vertical flexure modes and first two transverse flexure modes between the updated and damaged model. In MPSO process, the amount N and the maximum step were all 60, the inertia weight, learning coefficients and the mutation probability were chosen as 0.55, 2 and 0.1 respectively. In GA process, the mating probability and mutation probability were 0.8 and 0.2, the number of particles and maximum step were the same as before. The updating results of MPSO and GA were compared in Fig. 9 and the convergence process of each optimization method was shown in Fig. 10.

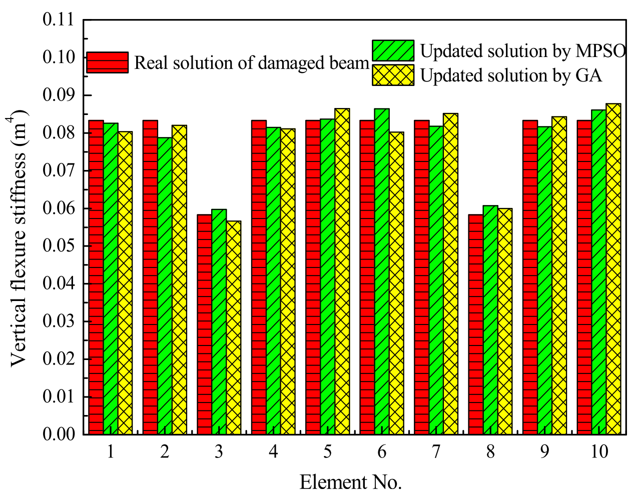


Fig. 9. The comparison between updated parameters of ten vertical flexure stiffnesses by MPSO and GA process.

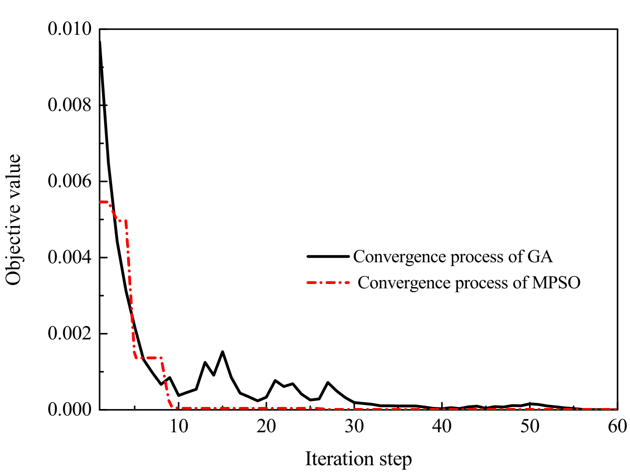


Fig. 10. Convergence process of MPSO and GA process.

In Fig. 9, the updated solutions of each parameter by MPSO and GA had good agreement with each other and the accuracy of the proposed model updating by MPSO is verified. Through the convergence process of MPSO and GA, the iteration step of MPSO and GA is 25 and 57 to reach the error of 1×10^{-5}. It is claimed that the convergence performance of MPSO is better than GA at the same error requirement. Hence, based on the model updating application of the simply supported beam example with 10 variables, the accuracy and convergence performance of the proposed method is feasible to be applied in model updating issue of engineering structures has high dimensional parameters.

4. Model updating of the extra-width bridge based on MPSO

4.1. Updating parameters and RSM expressions

In most FE model updating of bridge engineering structures, the dynamical characteristics of main girder are usually the interested responses considered in the objective functions. Although the dynamical characteristics of updated FE model shows good agreement with the testing results, the static responses are not well reflected. Herein, both the dynamical characteristics and the static responses of the typical points are considered in the objective functions. Through the sensitivity analysis to those responses of the bridge, several parameters are selected. The elastic modular of main girder x_1, elastic modular of main tower x_2, elastic modular of cable x_3, density of main girder x_4, vertical bending moment inertia of girder x_5, transverse bending moment inertia of girder x_6, torsion inertia of girder x_7, bending moment inertia of tower x_8, area of girder x_9 and secondary dead load x_{10} are sensitive to dynamical frequencies. The parameters of x_1 to x_7, x_9, x_{10} and the elastic modular of hanger x_{11}, area of tower x_{12} and temperature x_{13} are all sensitive to static displacements of typical points. The initial values of these parameters are listed in Table 5. Due to the high dimension of the parameters of 13 parameters, 200 input parameters are generated base on Latin Hypercube Sampling method and the values range from 60% to 140%. Then perform the initial 3-D FE model of this bridge 200 times and obtain 200 samples of the first five frequencies of the girder and vertical displacements of main girder. Based on the process of RSM, the surro-

Table 5
Updated values of parameters based on MPSO.

Parameters	Initial value	Updated value/ Initial value
Elastic modular of main girder x_1	3.45×10^{10} N/m^2	1.273780
Elastic modular of main tower x_2	3.25×10^{10} N/m^2	1.267193
Elastic modular of cable x_3	2.05×10^{11} N/m^2	1.059071
Density of main girder x_4	2.60×10^3 kg/m^3	1.023596
Vertical bending moment inertia of girder x_5	2.45–3.37 m^4	1.270247
Transverse bending moment inertia of girder x_6	5.11–42.90 m^4	1.250704
Torsion inertia of girder x_7	2.66–5.77 m^4	1.254238
Bending moment inertia of tower x_8	8.22 m^4	1.154252
Area of girder x_9	3.00–4.49 m^2	0.912652
Secondary dead load x_{10}	2.24×10^4–5.45×10^4 N/m^2	0.802291
Elastic modular of hanger x_{11}	2.05×10^{11} N/m^2	0.845129
Area of tower x_{12}	8.05 m^2	1.066491
Relative temperature x_{13}	273 K	1.104989

gate models of the frequencies and displacements from the mid-span of main-span and side-span are obtained. The coefficients in surrogate models are listed in Appendix A as supplementary data.

4.2. Model updating based on MPSO

On obtaining the second-order RSM expressions of responses, the objective function is determined. During the MPSO process, the objective function expressed in eq.(9)

$$F(\mathbf{X}_i) = \sum_{j=1}^{5}\left(f_j^{\text{test}} - f_j^{\text{RSM}}(\mathbf{X}_i)\right)^2 / \left(f_j^{\text{test}}\right)^2 + \sum_{i=1}^{2}\left(D_i^{\text{test}} - D_i^{\text{RSM}}(\mathbf{X}_i)\right)^2 / \left(D_i^{\text{test}}\right)^2 \quad (9)$$

is the summary of errors between the model and testing responses, where f denotes the mode frequency and D denotes the displacement of key point. The superscript 'test' and 'RSM' reflect the responses from in-situ experiment and former surrogate RSM models. Due to the surrogate RSM models are explicit relationships between responses and parameters, f_j^{RSM} and D_i^{RSM} are functions obtained ahead reflecting the relationships of frequencies and displacements. The physical meaning of this objective function is the coincident degree between the FE model and real structures. Through modifying the parameters $\mathbf{X}_i$, the purpose is yielding the minimum of $F(\mathbf{X}_i)$ and the corresponding $\mathbf{X}_i$. In the optimization process, the weighting factor is 0.55, acceleration coefficients are 2, swarm amount is 100, maximum step is 100, the stopping criteria of the objective function is less than 1×10^{-5} and the selecting probability in MPSO is still 0.1. The searching boundary of each parameter is 60–140% and the updated values of the parameters are then listed in Table 5.

Table 6
Comparisons of dynamical characteristics in the main girder.

Modes	Frequency (Hz)			Error (%)
	Updated	Testing	MAC	
First order of vertical bending	0.8672	0.9030	0.983	3.96
First order of torsion	1.3057	1.3670	0.972	4.48
First order of transverse bending	1.4745	1.3790	0.900	6.93
Second order of vertical bending	1.7373	1.9290	0.870	9.94
Third order of vertical bending	2.1760	2.4170	0.875	9.97

Note: Error = |Simulation-Test|/Simulation × 100%, MAC is modal assurance criteria.

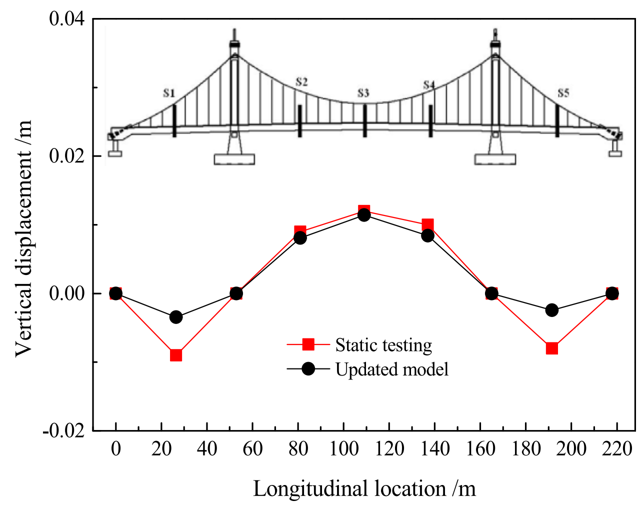


Fig. 12. Comparisons of modal shapes in the main girder.

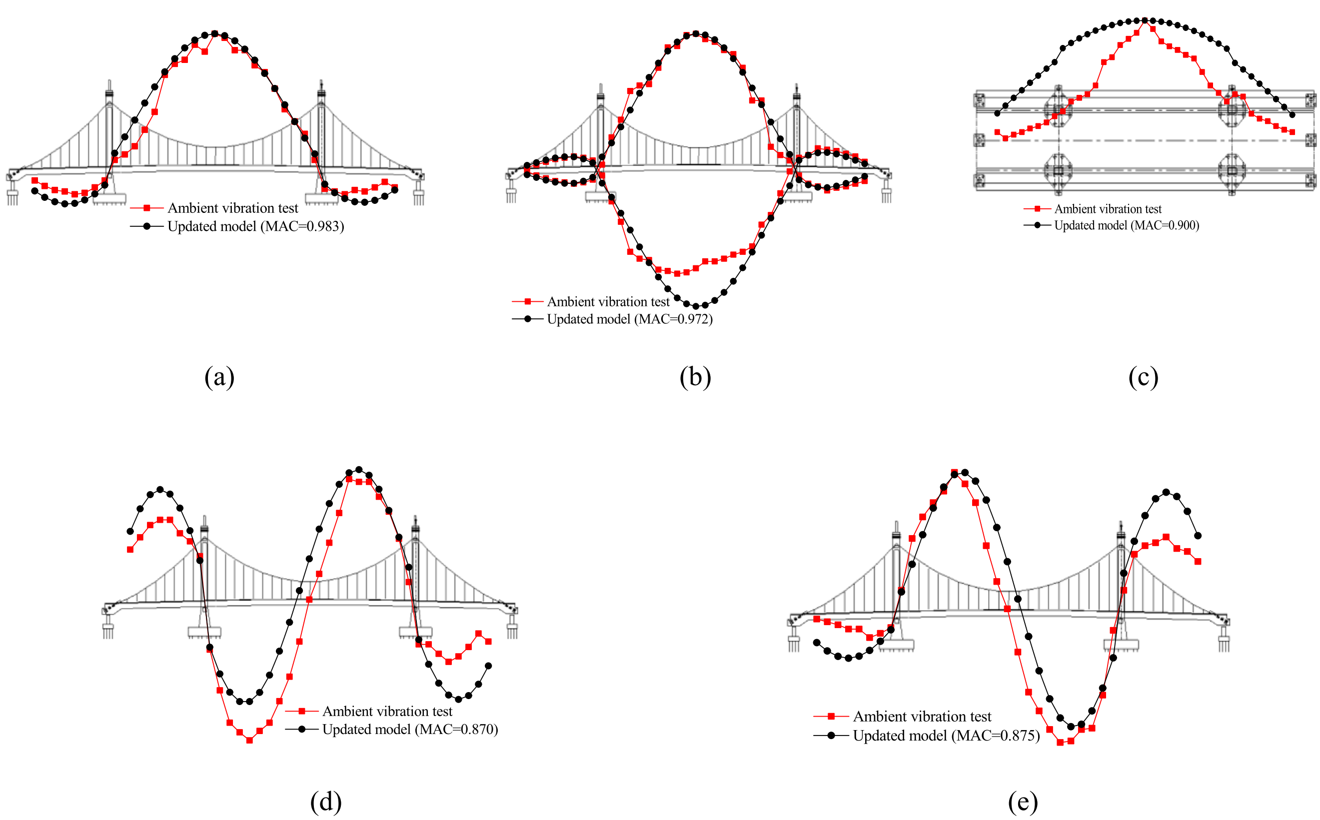


Fig. 11. Comparisons of modal shapes in the main girder: (a) First order of vertical bending, (b) First order of torsion, (c) First order of transverse bending, (d) Second order of vertical bending, and (e) Third order of vertical bending.

4.3. Validation of the updated model results

Perform the updated model of the bridge using the updated value of parameters, the first five frequencies, modal shapes and vertical displacements of typical points along the girder are all obtained. To illustrate the accuracy and feasibility of the MPSO applied in model updating of bridge engineering, the updated model results are compared with the testing data. The frequencies and MACs are compared in Table 6 and Fig. 11 respectively. The vertical displacements of mid-span of main span, quarter-span of main span and mid-span of side span are compared in Fig. 12. From the results comparisons, the first five frequencies of main girder are closer to the test values, where the errors are shortened from 40% to less than 10%. For the modal shapes of main girder, the MACs are all over 0.87, which means the shapes from FE model coincide with the experimental shapes well. Furthermore, except the dynamical characteristics of main girder, the vertical displacements of the typical points at the finished state also have good agreements. All the improvements in updated FE model reveal that the model updating based on MPSO and RSM is successful and the feasibility of MPSO applied in model updating of bridge engineering is available.

5. Conclusions

With the dimension of parameters in finite element (FE) model updating issues of engineering structures increasing, the traditional optimization methods are not immune to detecting global extremum. In this paper, a combined model updating method based on modified particle swarm optimization (PSO) and response surface method (RSM) was proposed. MPSO needs less modification to standard PSO and has faster convergence than PSO and genetic algorithm (GA) which has been widely used in bridge model, especially confronting with high dimensional parameters issues. After verification of numerical functions and engineering examples, this proposed technique was successfully applied to model updating of an existing bridge with thirteen parameters.

Based on the testing functions, the global extremum searching and convergence abilities of MPSO and PSO were compared. It is claimed that the global searching ability of MPSO is feasible and better than standard PSO even if dealing with high dimensional and multi-extremum complex issues.

With the comparison between the MPSO and GA method together with RSM applied to model updating of a simply supported beam with ten parameters, the convergence performance of MPSO is faster than GA at the same error requirement, as the iteration step of MPSO and GA is 25 and 57 to reach the error of 1×10^{-5}. Furthermore, the accuracy performance of the proposed method is acceptable in model updating issue with high dimensional parameters compared with the results of GA.

Then the application of the proposed method to model updating of an existing bridge structure with thirteen variables was successful. Through the comparisons between performances from updated FE model and in-situ experiment, it shows that the dynamical characteristics of girder and the static displacements of typical points from model are all closer to the test results, as the errors reduce from 40% to less than 10% and the shape of girder in vertical direction at the finished state is in good agreement with experimental shape.

CRediT authorship contribution statement

Zhiyuan Xia: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing - original draft, Writing - review & editing. **Aiqun Li:** Funding acquisition, Project administration, Resources, Supervision, Writing - review & editing. **Jianhui Li:** Investigation, Project administration, Resources, Software, Supervision. **Huiyuan Shi:** Data curation, Visualization, Writing - review & editing. **Maojun Duan:** Formal analysis, Validation, Visualization. **Guangpan Zhou:** Data curation, Funding acquisition, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was jointly supported by National Natural Science Foundation of China (Grant No. 51438002), the research fund of Jiangsu Province Key Laboratory of Structure Engineering, China (Grant No. ZD1803), Natural Science Foundation of Suzhou University of Science and Technology (Grant No. XKQ2018008), the Fundamental Research Funds for the Central Universities (Grant No. 30919011246), Natural Science Foundation of Jiangsu Higher Education Institutions of China (Grant Nos. 19KJB560021 and 19KJB560020) and a Project Funded by the Priority Academic Program Development of Jiangsu Higher Education Institutions.

Appendix A

This appendix provides the coefficients data of second-order polynomial RSM expressions of responses. f_1^{RSM} to f_5^{RSM} represent the frequencies in Table 2 or in Table 6 with the same sequence.

Table 7
Coefficients in surrogate models for first five frequencies.

CO	f_1^{RSM}	f_2^{RSM}	f_3^{RSM}	f_4^{RSM}	f_5^{RSM}	CO	f_1^{RSM}	f_2^{RSM}	f_3^{RSM}	f_4^{RSM}	f_5^{RSM}
β_0	0.5927	0.9288	1.2757	0.2342	0.9934	$\beta_{3,9}$	−0.0216	−0.0130	0.0564	−0.0715	−0.0291
$\beta_{1,1}$	−0.0609	−0.1170	−0.2144	−0.1236	−0.1090	$\beta_{3,10}$	−0.0101	−0.0111	−0.0174	0.0174	−0.0121
$\beta_{2,2}$	−0.0062	−0.0213	−0.0207	−0.0143	−0.0624	$\beta_{4,5}$	−0.0834	−0.0640	0.0282	−0.1152	−0.2206
$\beta_{3,3}$	−0.0389	−0.0504	0.0117	0.0601	−0.0226	$\beta_{4,6}$	0.0000	0.0050	0.0302	−0.0356	0.0045
$\beta_{4,4}$	0.1728	0.2725	0.4202	0.3138	0.3594	$\beta_{4,7}$	−0.0027	−0.0606	0.0089	0.0758	−0.0111
$\beta_{5,5}$	−0.0323	−0.0104	0.0249	−0.0606	−0.0090	$\beta_{4,8}$	−0.0122	−0.0180	0.0018	−0.0200	−0.0480
$\beta_{6,6}$	0.0024	−0.0075	−0.0407	−0.1334	0.0070	$\beta_{4,9}$	0.0244	0.0549	−0.0257	0.1451	0.0185
$\beta_{7,7}$	−0.0036	−0.0326	−0.0219	−0.0555	−0.0146	$\beta_{4,10}$	0.1008	0.1405	0.2013	−0.0052	0.2162
$\beta_{8,8}$	−0.0053	−0.0138	0.0008	0.0043	−0.0870	$\beta_{5,6}$	0.0040	0.0029	0.0106	−0.0735	−0.0070
$\beta_{9,9}$	0.0534	0.0832	−0.1478	0.0869	0.0827	$\beta_{5,7}$	0.0013	−0.0286	−0.0287	−0.1400	−0.0068
$\beta_{10,10}$	0.0124	0.0161	−0.0547	0.0303	0.0459	$\beta_{5,8}$	0.0011	0.0048	0.0339	−0.1410	−0.1162
$\beta_{1,2}$	−0.0037	0.0072	0.0010	−0.0395	−0.0550	$\beta_{5,9}$	−0.0598	−0.0223	−0.0196	−0.0590	−0.1553

(continued on next page)

Table 7 (*continued*)

CO	f_1^{RSM}	f_2^{RSM}	f_3^{RSM}	f_4^{RSM}	f_5^{RSM}	CO	f_1^{RSM}	f_2^{RSM}	f_3^{RSM}	f_4^{RSM}	f_5^{RSM}
$\beta_{1,3}$	−0.0066	−0.0218	0.0003	0.0117	−0.0368	$\beta_{5,10}$	−0.0262	−0.0167	−0.0137	−0.0085	−0.0075
$\beta_{1,4}$	−0.1016	−0.1501	−0.3386	−0.1389	−0.2603	$\beta_{6,7}$	0.0019	0.0020	0.0180	−0.0070	0.0060
$\beta_{1,5}$	0.1122	0.1070	−0.0074	0.2563	0.3734	$\beta_{6,8}$	0.0005	−0.0030	−0.0053	0.0880	−0.0045
$\beta_{1,6}$	−0.0041	−0.0078	0.0276	0.0686	−0.0081	$\beta_{6,9}$	−0.0069	−0.0059	0.0344	0.0156	−0.0049
$\beta_{1,7}$	0.0033	0.1024	0.0448	−0.1498	−0.0033	$\beta_{6,10}$	0.0013	0.0127	0.0161	0.0099	−0.0020
$\beta_{1,8}$	−0.0040	0.0060	0.0321	−0.1430	−0.0581	$\beta_{7,8}$	−0.0014	−0.0044	−0.0032	0.1323	0.0276
$\beta_{1,9}$	−0.0592	−0.1611	−0.0528	−0.0792	−0.2362	$\beta_{7,9}$	−0.0002	−0.0249	0.0184	0.0292	−0.0014
$\beta_{1,10}$	−0.0367	−0.0551	−0.1105	−0.0046	0.0006	$\beta_{7,10}$	0.0042	0.0009	0.0220	−0.0535	0.0026
$\beta_{2,3}$	0.0026	−0.0032	−0.0148	−0.0314	0.0373	$\beta_{8,9}$	−0.0060	0.0105	0.0043	−0.0742	0.0534
$\beta_{2,4}$	−0.0081	−0.0202	−0.0238	−0.2796	−0.0575	$\beta_{8,10}$	−0.0018	−0.0010	0.0081	−0.1367	−0.0742
$B_{2,5}$	−0.0040	−0.0143	0.0369	−0.1609	−0.1589	$\beta_{9,10}$	0.0619	0.1024	0.0768	−0.0654	0.1405
$\beta_{2,6}$	0.0016	0.0024	0.0257	−0.0508	−0.0130	β_1	0.4661	0.8150	1.4984	1.0278	1.0674
$\beta_{2,7}$	0.0050	0.0089	−0.0053	0.1310	0.0038	β_2	0.0473	0.0980	0.1009	0.3979	0.5821
$\beta_{2,8}$	0.0053	0.0000	−0.0276	0.2284	0.1259	β_3	0.2571	0.3716	−0.0330	0.0227	0.1671
$\beta_{2,9}$	−0.0111	−0.0002	−0.0194	−0.1192	0.0349	β_4	−0.5089	−0.8330	−1.2897	−0.7026	−0.9900
$\beta_{2,10}$	−0.0005	−0.0050	−0.0008	0.0687	−0.0727	β_5	0.3223	0.2325	−0.0609	0.8729	0.7301
$\beta_{3,4}$	−0.0420	−0.0480	0.0090	−0.0352	−0.0136	β_6	−0.0015	0.0134	−0.0817	0.3109	0.0043
$\beta_{3,5}$	−0.0119	−0.0383	−0.0203	0.1261	−0.0384	β_7	−0.0049	0.2187	−0.0197	−0.0212	0.0085
$\beta_{3,6}$	−0.0028	−0.0051	0.0106	−0.0598	0.0082	β_8	0.0452	0.0554	−0.0146	0.2349	0.4961
$\beta_{3,7}$	0.0001	−0.0135	−0.0235	0.0760	0.0028	β_9	−0.1977	−0.3907	0.2421	−0.1605	−0.3362
$\beta_{3,8}$	0.0005	−0.0023	−0.0076	−0.1225	−0.0062	β_{10}	−0.1936	−0.3125	−0.2101	−0.0158	−0.5034

Table 8
Coefficients in surrogate models for vertical displacements.

CO	D_1^{RSM}	D_2^{RSM}	CO	D_1^{RSM}	D_2^{RSM}	CO	D_1^{RSM}	D_2^{RSM}
β_0	0.4113	0.3596	$\beta_{2,11}$	0.0017	0.0020	$\beta_{6,12}$	−0.0107	0.0021
$\beta_{1,1}$	−0.0484	0.0290	$\beta_{2,12}$	−0.0363	0.0009	$\beta_{6,13}$	−0.0142	0.0041
$\beta_{2,2}$	−0.0194	−0.0034	$\beta_{2,13}$	−0.0023	0.0111	$\beta_{7,9}$	−0.0152	0.0036
$\beta_{3,3}$	−0.1812	0.0121	$\beta_{3,4}$	0.1142	−0.0017	$\beta_{7,10}$	0.0057	0.0002
$\beta_{4,4}$	0.0042	0.0049	$\beta_{3,5}$	−0.1317	0.0076	$\beta_{7,11}$	0.0087	−0.0029
$\beta_{5,5}$	−0.0145	0.0105	$\beta_{3,6}$	0.0053	0.0004	$\beta_{7,12}$	0.0047	−0.0052
$\beta_{6,6}$	0.0172	−0.0045	$\beta_{3,7}$	0.0193	0.0009	$\beta_{7,13}$	0.0131	0.0088
$\beta_{7,7}$	−0.0006	0.0049	$\beta_{3,9}$	0.0940	−0.0050	$\beta_{9,10}$	0.0040	0.0008
$\beta_{9,9}$	−0.0096	0.0049	$\beta_{3,10}$	0.0401	−0.0009	$\beta_{9,11}$	0.0082	0.0001
$\beta_{10,10}$	−0.0035	0.0007	$\beta_{3,11}$	0.0542	−0.0008	$\beta_{9,12}$	0.0107	−0.0016
$\beta_{11,11}$	−0.0590	0.0042	$\beta_{3,12}$	0.0114	0.0030	$\beta_{9,13}$	0.0936	−0.0279
$\beta_{12,12}$	−0.0479	0.0033	$\beta_{3,13}$	−0.0968	0.0058	$\beta_{10,11}$	0.0215	−0.0033
$\beta_{13,13}$	−0.0230	−0.0028	$\beta_{4,5}$	0.1507	−0.0027	$\beta_{10,12}$	−0.0246	−0.0007
$\beta_{1,2}$	0.0031	0.0014	$\beta_{4,6}$	−0.0032	0.0009	$\beta_{10,13}$	0.0095	−0.0004
$\beta_{1,3}$	−0.1942	0.0148	$\beta_{4,7}$	0.0152	0.0005	$\beta_{11,12}$	0.0248	−0.0015
$\beta_{1,4}$	0.1852	−0.0002	$\beta_{4,9}$	−0.2210	−0.0015	$\beta_{11,13}$	−0.0277	0.0037
$\beta_{1,5}$	0.0022	0.0157	$\beta_{4,10}$	0.0000	−0.0038	$\beta_{12,13}$	−0.0117	0.0067
$\beta_{1,6}$	0.0169	−0.0001	$\beta_{4,11}$	0.0422	−0.0060	β_1	−0.0682	−0.0946
$\beta_{1,7}$	0.0098	0.0062	$\beta_{4,12}$	0.0041	0.0025	β_2	0.0414	−0.0057
$\beta_{1,9}$	0.0837	0.0124	$\beta_{4,13}$	0.0049	−0.0017	β_3	0.7947	−0.0718
$\beta_{1,10}$	0.0496	−0.0089	$\beta_{5,6}$	0.0004	0.0002	β_4	−0.6750	0.0098
$\beta_{1,11}$	−0.0353	−0.0012	$\beta_{5,7}$	0.0202	0.0016	β_5	−0.0505	−0.1395
$\beta_{1,12}$	−0.0125	−0.0011	$\beta_{5,9}$	0.0599	0.0080	β_6	−0.0198	0.0077
$\beta_{1,13}$	0.1012	−0.0261	$\beta_{5,10}$	0.0411	−0.0095	β_7	−0.0830	−0.0209
$\beta_{2,3}$	0.0282	−0.0015	$\beta_{5,11}$	−0.0235	−0.0038	β_9	−0.3232	−0.0045
$\beta_{2,4}$	0.0153	0.0019	$\beta_{5,12}$	−0.0195	0.0040	β_{10}	−0.2245	0.0381
$B_{2,5}$	−0.0016	0.0009	$\beta_{5,13}$	0.0048	0.0835	β_{11}	0.1115	0.0028
$\beta_{2,6}$	0.0106	−0.0029	$\beta_{6,7}$	−0.0039	−0.0028	β_{12}	0.1806	−0.0143
$\beta_{2,7}$	−0.0027	0.0002	$\beta_{6,9}$	−0.0005	−0.0011	β_{13}	−0.2125	−0.2044
$\beta_{2,9}$	0.0090	−0.0002	$\beta_{6,10}$	−0.0083	0.0018			
$\beta_{2,10}$	0.0046	−0.0026	$\beta_{6,11}$	−0.0033	−0.0001			

D_1^{RSM} and D_2^{RSM} are vertical displacements of points at mid-span of main span and mid-span of side span respectively. The meaning of the coefficients (CO) refers to Eq. (6). The frequencies and displacements coefficients are listed in Tables 7 and 8.

References

[1] M. Mashayekhi, E. Santini-Bell, Three-dimensional multiscale finite element models for in-service performance assessment of bridges, Comput.-Aided. Civ. Inf. 34 (2019) 385–401.

[2] Y.Z. Liang, F. Xiong, Measurement-based bearing capacity evaluation for small and medium span bridges, Measurement 149 (2020), UNSP 106938.

[3] Z.Y. Xia, S.T. Quek, A.Q. Li, J.H. Li, M.J. Duan, Hybrid approach to seismic reliability assessment of engineering structures, Eng. Struct. 153 (2017) 665–673.

[4] D.H. Yang, T.H. Yi, H.N. Li, Y.F. Zhang, Monitoring and analysis of thermal effect on tower displacement in cable-stayed bridge, Measurement 115 (2018) 249–257.

[5] T.H. Yi, H.N. Li, M. Gu, Effect of different construction materials on propagation of GPS monitoring signals, Measurement 45 (2012) 1126–1139.

[6] T. Guo, Z.H. Chen, S. Lu, R.G. Yao, Monitoring and analysis of long-term prestress losses in post-tensioned concrete beams, Measurement 122 (2018) 573–581.

[7] Y.S. Erdogan, M. Gul, F.N. Catbas, P.G. Bakir, Investigation of uncertainty changes in model outputs for finite-element model updating using structural health monitoring data, J. Struct. Eng. 140 (2014), 04014078.

[8] D.M. Feng, M.Q. Feng, Model updating of railway bridge using in situ dynamic displacement measurement under trainloads, J. Bridge Eng. 20 (2015), 04015019.

[9] A.C. Altunisik, E. Kalkan, F.Y. Okur, K. Ozgan, O.S. Karahasan, A. Bostanci, Non-destructive modal parameter identification of historical timber bridges using ambient vibration tests after restoration, Measurement 146 (2019) 411–424.

[10] A.C. Altunişik, O.Ş. Karahasan, A.F. Genç, F.Y. Okur, M. Günaydin, E. Kalkan, S. Adanur, Modal parameter identification of RC frame under undamaged, damaged, repaired and strengthened conditions, Measurement 124 (2018) 260–276.
[11] H.S. Park, J.H. Kim, B.K. Oh, Model updating method for damage detection of building structures under ambient excitation using modal participation ratio, Measurement 133 (2019) 251–261.
[12] W.X. Ren, S.E. Fang, M.Y. Deng, Response surface-based finite-element-model updating using structural static responses, J. Eng. Mech. 137 (2011) 248–257.
[13] W.X. Ren, H.B. Chen, Finite element model updating in structural dynamics by using response surface method, Eng. Struct. 32 (2010) 2455–2465.
[14] L. Deng, C.S. Cai, Bridge model updating using response surface method and genetic algorithm, J. Bridge Eng. 15 (2010) 553–564.
[15] H.P. Wan, W.X. Ren, Parameter selection in finite-element-model updating by global sensitivity analysis using Gaussian process metamodel, J. Struct. Eng. 141 (2015), 04014164.
[16] S.Q. Qin, Y.L. Zhou, H.Y. Cao, M.A. Wahab, Model updating in complex bridge structures using Kriging model ensemble with genetic algorithm, KSCE J. Civ. Eng. 22 (2018) 3567–3578.
[17] F. Shabbir, P. Omenzetter, Model updating using genetic algorithms with sequential niche technique, Eng. Struct. 120 (2016) 166–182.
[18] S.Q. Qin, J. Hu, Y.L. Zhou, Y.Z. Zhang, J.T. Kang, Feasibility study of improved particle swarm optimization in Kriging metamodel based structural model updating, Struct. Eng. Mech. 70 (2019) 513–524.
[19] R. Perera, S.E. Fang, A. Ruiz, Application of particle swarm optimization and genetic algorithms to multiobjective damage identification inverse problems with modelling errors, Meccanica 45 (2010) 723–734.
[20] N.M. Okasha, D.M. Frangopol, A.D. Orcesi, Automated finite element updating using strain data for the lifetime reliability assessment of bridges, Reliab. Eng. Syst. Saf. 99 (2012) 139–150.
[21] M. Vahidi, S. Vandani, M. Rahimian, N. Jamshidi, A.T. Kanee, Evolutionary-base finite element model updating and damage detection using modal testing results, Struct. Eng. Mech. 70 (2019) 339–350.
[22] Y.K. Li, Y.L. Chen, J.H. Zhong, Z.X. Huang, Niching particle swarm optimization with equilibrium factor for multi-modal optimization, Inf. Sci. 494 (2019) 233–246.
[23] F. Shabbir, P. Omenzetter, Particle swarm optimization with sequential niche technique for dynamic finite element model updating, Comput.-Aided Civ. Infrastruct. Eng. 30 (2015) 359–375.
[24] B. Nanda, D. Maity, D.K. Maiti, Crack assessment in frame structures using modal data and unified particle swarm optimization technique, Adv. Struct. Eng. 17 (2014) 747–766.
[25] S. Gholizadeh, R.K. Moghadas, Performance-based optimum design of steel frames by an improved quantum particle swarm optimization, Adv. Struct. Eng. 17 (2014) 143–156.
[26] N. Higashi, H. Iba, Particle swarm optimization with Gaussian mutation, 2003, Proceedings of the 2003 IEEE, Swarm Intelligence Symposium, 72–79.
[27] R.J. Xi, W.P. Jiang, X.L. Meng, H. Chen, Q.S. Chen, Bridge monitoring using BDS-RTK and GPS-RTK techniques, Measurement 120 (2018) 128–139.
[28] J.Y. Yu, P. Zhu, B. Xu, X.L. Meng, Experimental assessment of high sampling-rate robotic total station for monitoring bridge dynamic responses, Measurement 104 (2017) 60–69.
[29] T.H. Yi, H.N. Li, M. Gu, Experimental assessment of high-rate GPS receivers for deformation monitoring of bridge, Measurement 46 (2013) 420–432.
[30] A. Shariati, T. Schumacher, Eulerian-based virtual visual sensors to measure dynamic displacements of structures, Struct. Control Health 24 (2017) e1977.
[31] S. Umar, N. Bakhary, A.R.Z. Abidin, Response surface methodology for damage detection using frequency and mode shape, Measurement 115 (2018) 258–268.
[32] H. Dette, A. Pepelyshev, Generalized Latin Hypercube Design for computer experiments, Technometrics 52 (2010) 421–429.